Is "best pack" actually the best way to fill the binder?

A Monte-Carlo comparison of three pack-buying strategies for the National Pokédex project

Generated 1 June 2026 · 2,000 simulated collectors per strategy · 1,000 packs each · seed 1234

The question

The tracker's dashboard recommends a "best pack" — always open a booster from the set with the highest *expected number of new Pokémon species*, given what you already own. Is that genuinely better than buying blindly, or does it just feel clever? We pit it against two alternatives:

Strategy A

Best pack

Always open the set with the highest expected new-species yield, recomputed as the collection grows. (What the app recommends.)

Strategy B

Random set

Open a pack from a uniformly random set every time. The "no strategy" baseline.

Strategy C

Least-opened

Open a pack from whichever set you've opened fewest so far (ties broken at random). A balanced round-robin that ignores card value.

Each strategy starts from an **empty collection** and we measure the average number of distinct species collected after 10, 50, 200 and 1,000 packs.

Verdict

Best pack wins — clearly, and at every checkpoint. It is never worse than the alternatives, and the smarter the choice set, the bigger its lead.

Across the project's own Scarlet & Violet + Mega Evolution sets, after 200 packs it collects about **724 species — 32 more than random**. Given the full TCG catalogue to choose from, that lead balloons to **124 species**. Random and least-opened are nearly tied, with least-opened a hair ahead.

And the bill to *finish* the dex (last sections): about **€11,398** opening best packs vs **€110,595** buying at random — or as little as **€1,014** if you buy the hard cards as singles (Rares included) instead of chasing them in packs.

At a steady **one booster + ten cards a week**, the binder fills in about **72 weeks (~1.4 years)** — and **trading your duplicate commons** for cards you need covers ~301 of them, shaving roughly **€150** off the bill without changing the timeline.

How the simulation works

- **Real pack model.** Packs are opened with the app's own booster simulator (`lib/packs/simulator.ts`): era-specific slot layouts and community-sourced rare-slot pull weights, drawing from each set's actual card pool by rarity. Only species you can really pull are counted.
- **Exact "best pack" decisions.** Instead of re-running Monte-Carlo each step, the best-pack strategy uses a closed-form expected-new value computed from the same pull probabilities. We verified it matches the app's 5,000-iteration simulator to within ~0.01 species across every era (it is the same quantity, computed exactly rather than estimated).

- **Honest ceilings.** Not every listed card is pullable from boosters (secret rares, promos). Curves are measured against the *achievable* ceiling — the species actually reachable by opening packs.
- **Statistics.** 2,000 independent collectors per strategy; we report the mean with a 95% confidence interval (normal and percentile-bootstrap agree). With this many trials the intervals are a fraction of a species wide — far narrower than the gaps between strategies.
- **Reproducible & independent.** Every collector is driven by a seeded RNG derived from one base seed. We use independent streams per strategy rather than common random numbers: the strategies open different sets each step, so their draw streams desynchronise immediately and shared seeds would buy no variance reduction here.

Scarlet & Violet + Mega Evolution

21 openable sets · 933 species are actually pullable from packs (the achievable ceiling), out of 1025 in the National Pokédex.

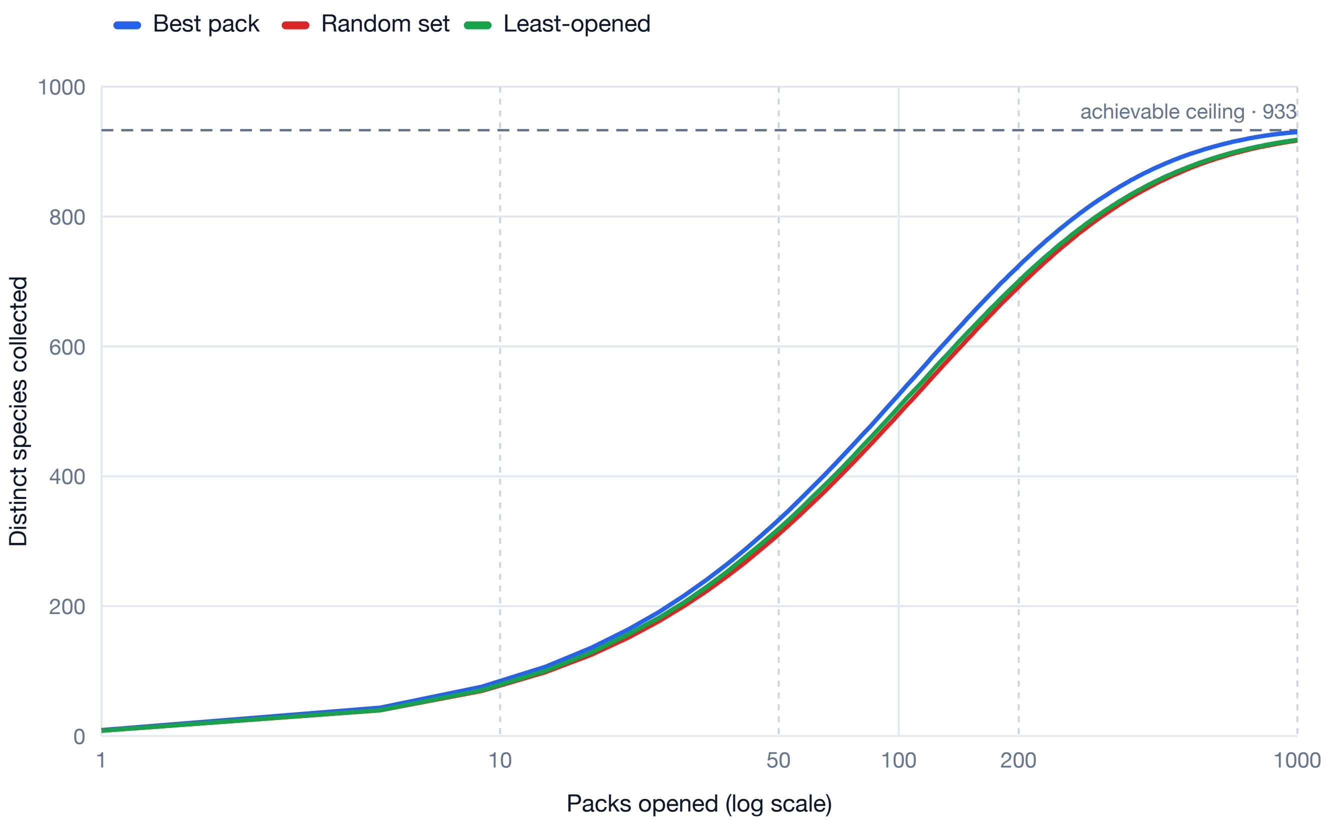

Figure 1. Mean distinct species collected as packs are opened, averaged over 2,000 simulated collectors per strategy. Shaded bands are 95% confidence intervals (so narrow they are barely visible). The dashed line is the achievable ceiling.

After **200 packs**, the best-pack collector holds about **724 species** — roughly **32 more** than random buying (95% CI 31.4–32.8; the intervals are far too narrow to overlap, so the difference is unambiguous).

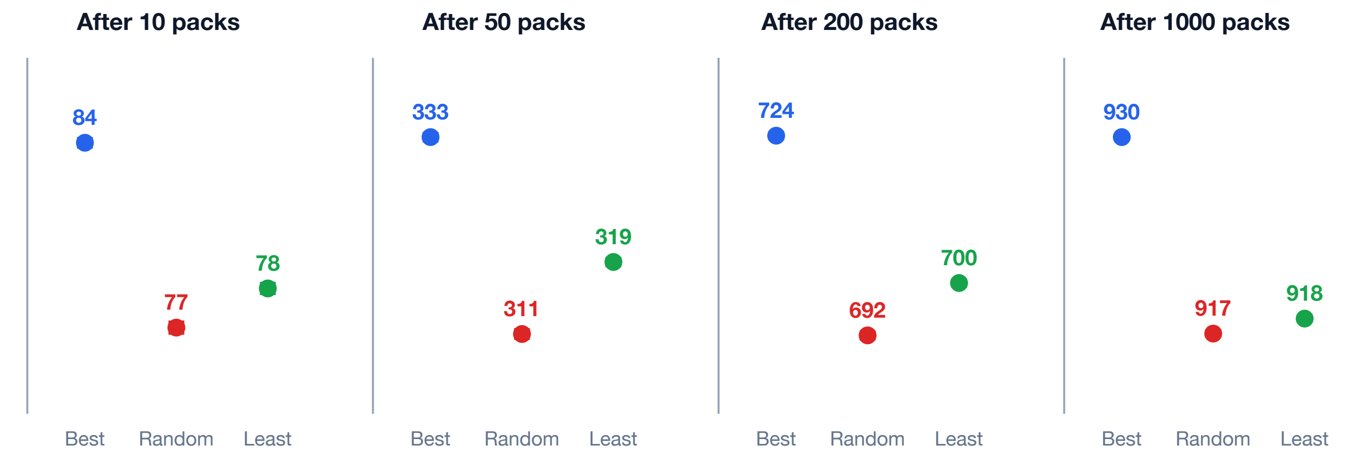

Figure 2. The four checkpoints at zoomed scale: dots are means, whiskers are 95% CIs. Each panel is scaled to its own range so the gaps are visible.

Mean species collected

Strategy	After 10 packs	After 50 packs	After 200 packs	After 1000 packs
Best pack	**83.9** 83.7–84.0	**332.9** 332.6–333.3	**724.1** 723.7–724.6	**930.1** 930.0–930.2
Random set	**76.8** 76.7–77.0	**310.9** 310.5–311.4	**692.0** 691.5–692.6	**917.2** 917.0–917.3
Least-opened	**78.3** 78.2–78.5	**319.0** 318.6–319.4	**700.5** 700.0–700.9	**918.2** 918.0–918.3

How far ahead is "best pack"?

Best pack advantage	After 10 packs	After 50 packs	After 200 packs	After 1000 packs
vs Random set	**+7.0** 95% CI 6.8–7.3	**+22.0** 95% CI 21.4–22.6	**+32.1** 95% CI 31.4–32.8	**+13.0** 95% CI 12.8–13.2
vs Least-opened	**+5.5** 95% CI 5.3–5.8	**+14.0** 95% CI 13.4–14.5	**+23.7** 95% CI 23.0–24.3	**+12.0** 95% CI 11.8–12.2

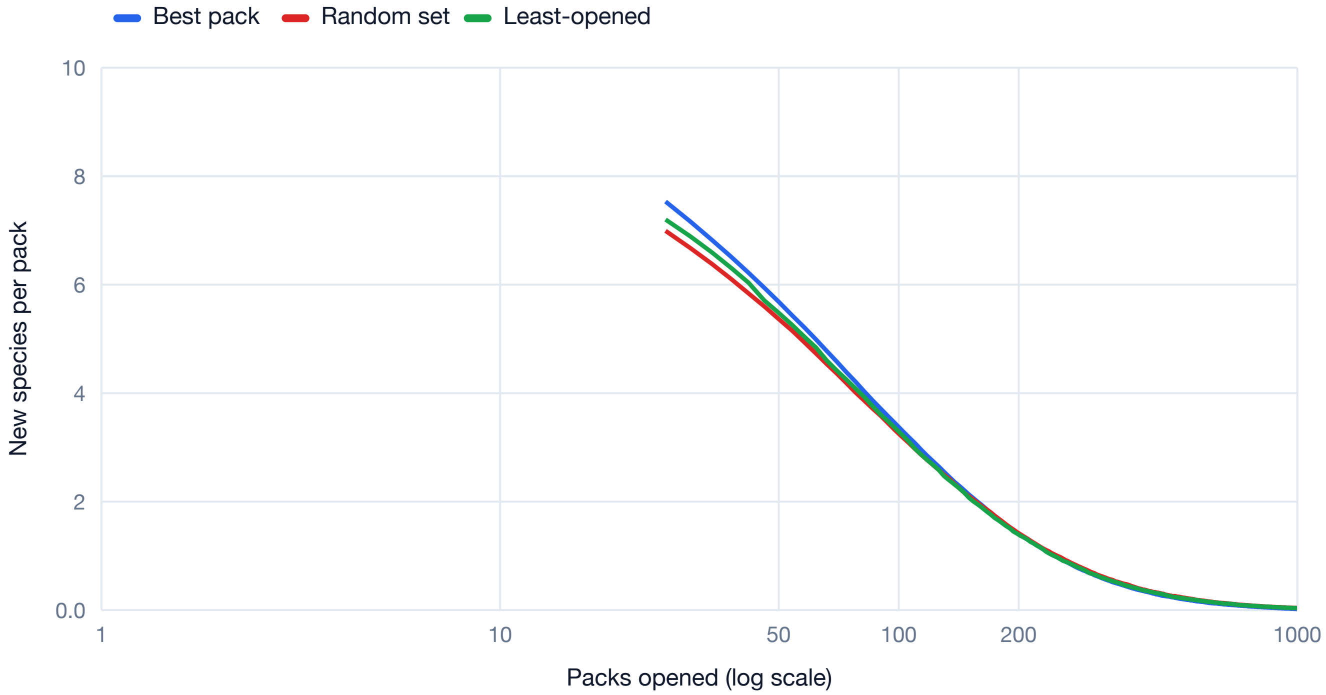

Figure 3. Marginal yield — new species gained per pack (smoothed). Best-pack stays higher for longer; all strategies decay toward zero as the collection saturates.

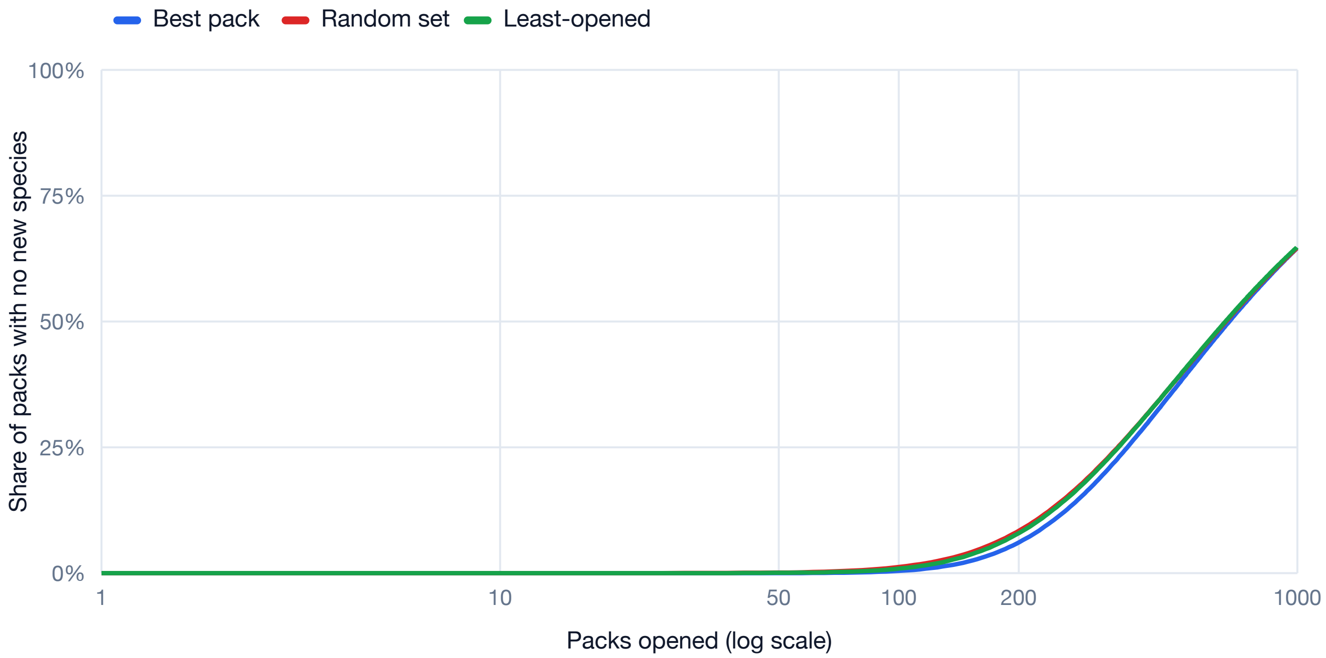

Figure 4. Share of packs that yielded *no* new species. Random and least-opened waste packs much sooner because they keep buying sets whose Pokémon you already have.

By **1000 packs** best-pack reaches **930** of the 933 achievable species, versus **917** for random — and it gets there having wasted far fewer packs along the way.

What you actually pull

Following the best-pack strategy for 1,000 packs (averaged over 1,000 runs), you open about **10000 cards**: **8497 Pokémon** (85%), 1433 Trainers and 70 Energy. Those Pokémon cards resolve to **930 distinct species**, which means roughly **89% of the Pokémon cards are duplicates** of species you already had — the price of chasing the long tail.

Cards by rarity (cumulative, average)

Rarity	@ 10	@ 50	@ 200	@ 1000
Common	50	251	1004	5018
Uncommon	38	187	747	3727
Rare	8.3	41	168	851
Double Rare	2.2	11	43	213
Ultra Rare	0.7	3.5	14	71
Illustration Rare	0.8	4.2	16	79
Special Illus. Rare	0.3	1.6	6.6	33
Hyper Rare	0.1	0.5	1.7	9.0
All cards	100	500	2000	10000

Commons and Uncommons dominate every pack; the higher tiers come almost entirely from the single guaranteed "rare slot" (and the occasional reverse-holo).

What lands in the rare slot?

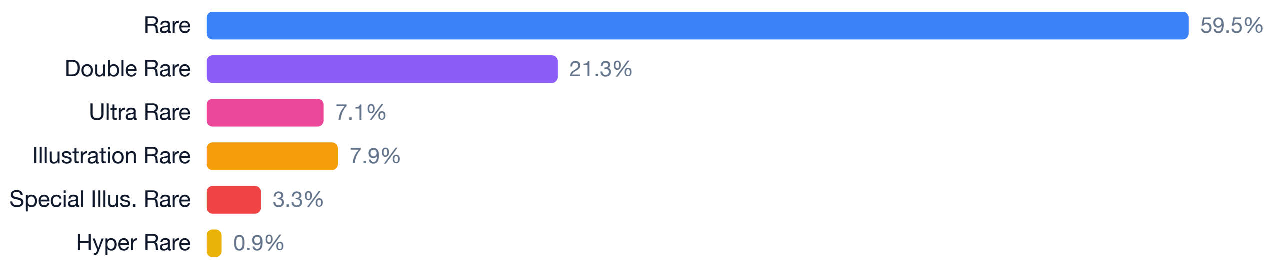

Figure 5. Of the one guaranteed rare-slot card per pack, how often each tier appears. These broadly follow the Scarlet & Violet pull weights; sets that lack a given chase tier route that slot to a plain Rare, lifting Rare a few points above its nominal 55%.

The most-pulled Pokémon

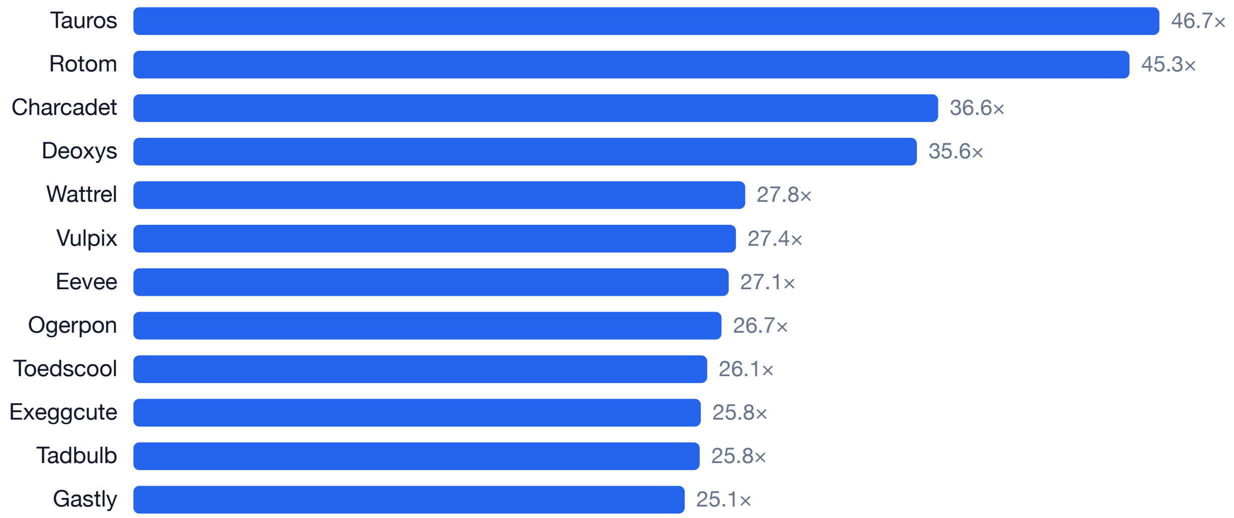

Figure 6. Average copies collected in 1,000 packs. These are common Pokémon reprinted across many of the sets best-pack opens — you end up swimming in them.

The hardest to pull

The 15 scarcest species that still showed up at least once — typically found only at high rarity or in sets best-pack rarely needs to open (every one of the 933 reachable species turned up at least once).

Species	Gen	Copies	Species	Gen	Copies
Lugia	Johto	0.25	Dragalge	Kalos	1.77
Empoleon	Sinnoh	0.92	Dragapult	Galar	1.79
Torterra	Sinnoh	1.11	Jellicent	Unova	1.83
Incineroar	Alola	1.15	Jumpluff	Johto	1.90
Zygarde	Kalos	1.22	Mime Jr.	Sinnoh	1.91
Yanmega	Sinnoh	1.29	Cacnea	Hoenn	2.09
Gallade	Sinnoh	1.30	Baxcalibur	Paldea	2.09
Salamence	Hoenn	1.33			

Why the lead grows with more sets

The size of best-pack's advantage depends on **how much choice it has**. With only the 21 Scarlet & Violet + Mega sets — which share many of the same popular Pokémon — random buying stumbles into those species anyway, so the gap is modest. Open the whole TCG catalogue (133 sets spanning every era) and best-pack can keep steering toward sets full of species you're missing, while random and least-opened squander packs on overlapping reprints. That is exactly what the appendix below shows: the 200-pack lead jumps from ~32 to ~124 species.

A secondary lesson: **least-opened modestly beats pure random** in most cases. Spreading purchases evenly across sets is a cheap way to avoid over-investing in one set you've already drained — but it is no substitute for actually targeting your gaps.

Appendix — All openable sets (Base → today)

133 openable sets · 1025 species are actually pullable from packs (the achievable ceiling), out of 1025 in the National Pokédex.

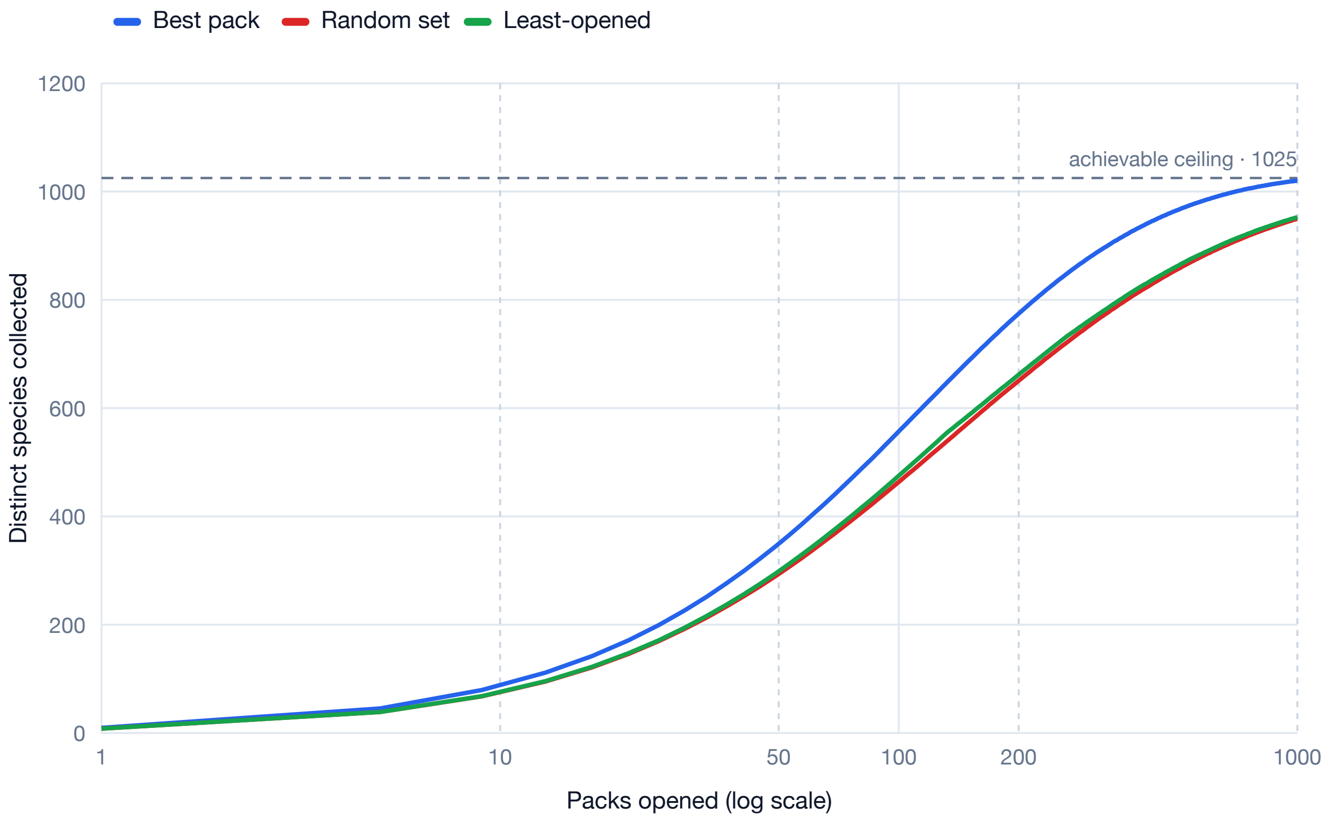

Figure A1. Mean distinct species collected as packs are opened, averaged over 2,000 simulated collectors per strategy. Shaded bands are 95% confidence intervals (so narrow they are barely visible). The dashed line is the achievable ceiling.

After **200 packs**, the best-pack collector holds about **775 species** — roughly **124 more** than random buying (95% CI 123.1–124.7; the intervals are far too narrow to overlap, so the difference is unambiguous).

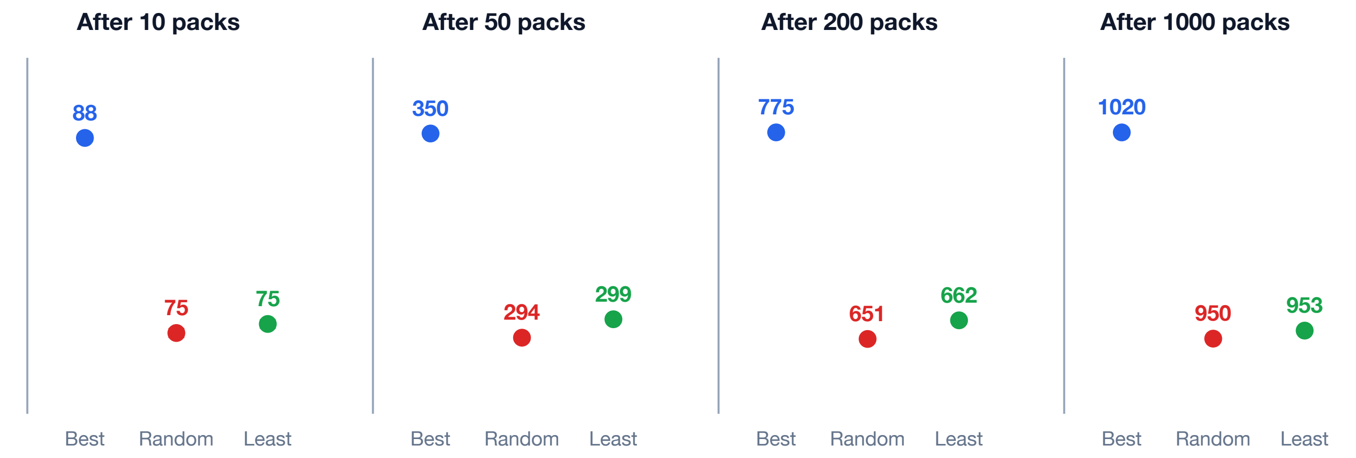

Figure A2. The four checkpoints at zoomed scale: dots are means, whiskers are 95% CIs. Each panel is scaled to its own range so the gaps are visible.

Mean species collected

Strategy	After 10 packs	After 50 packs	After 200 packs	After 1000 packs
Best pack	**87.5** 87.4–87.7	**349.6** 349.3–350.0	**774.7** 774.2–775.1	**1020.1** 1019.9–1020.2
Random set	**74.6** 74.4–74.8	**293.7** 293.2–294.2	**650.8** 650.1–651.4	**949.9** 949.5–950.2
Least-opened	**75.2** 75.0–75.4	**298.8** 298.3–299.2	**662.0** 661.5–662.5	**952.6** 952.3–952.9

How far ahead is "best pack"?

Best pack advantage	After 10 packs	After 50 packs	After 200 packs	After 1000 packs
vs Random set	**+13.0** 95% CI 12.7–13.2	**+55.9** 95% CI 55.3–56.5	**+123.9** 95% CI 123.1–124.7	**+70.2** 95% CI 69.8–70.6
vs Least-opened	**+12.4** 95% CI 12.1–12.6	**+50.9** 95% CI 50.3–51.4	**+112.7** 95% CI 112.1–113.4	**+67.5** 95% CI 67.2–67.8

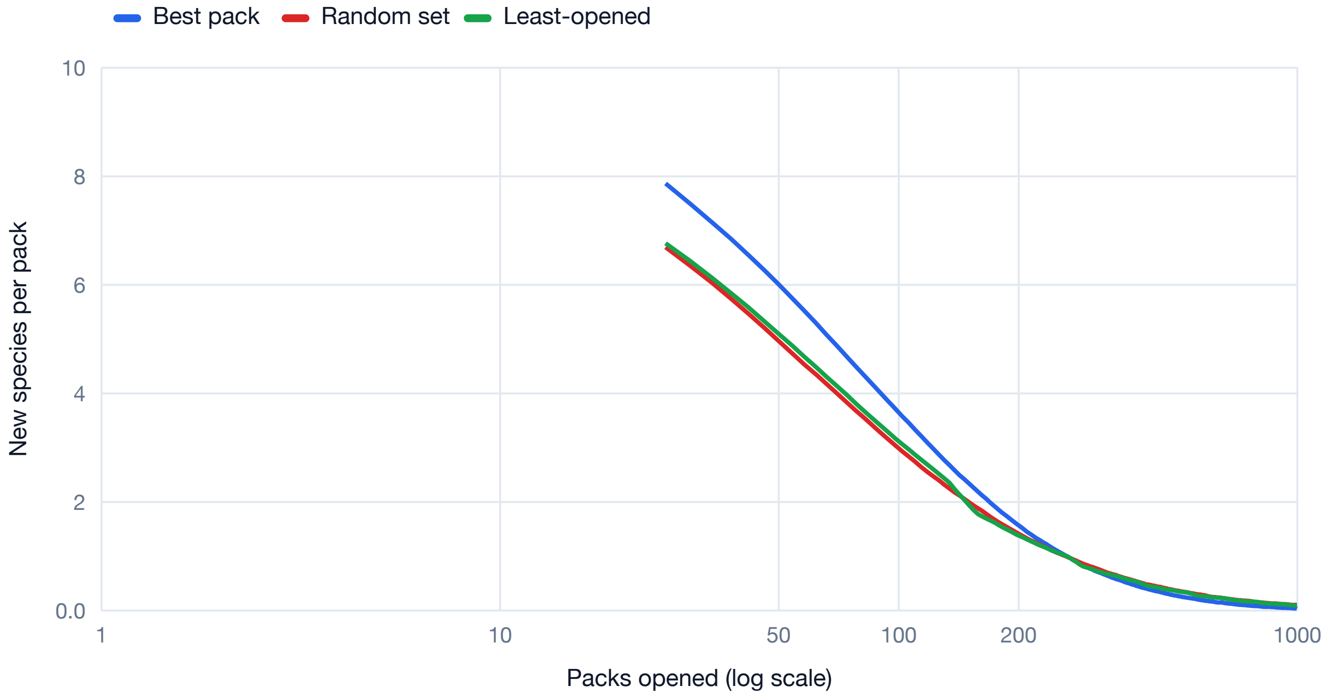

Figure A3. Marginal yield — new species gained per pack (smoothed). Best-pack stays higher for longer; all strategies decay toward zero as the collection saturates.

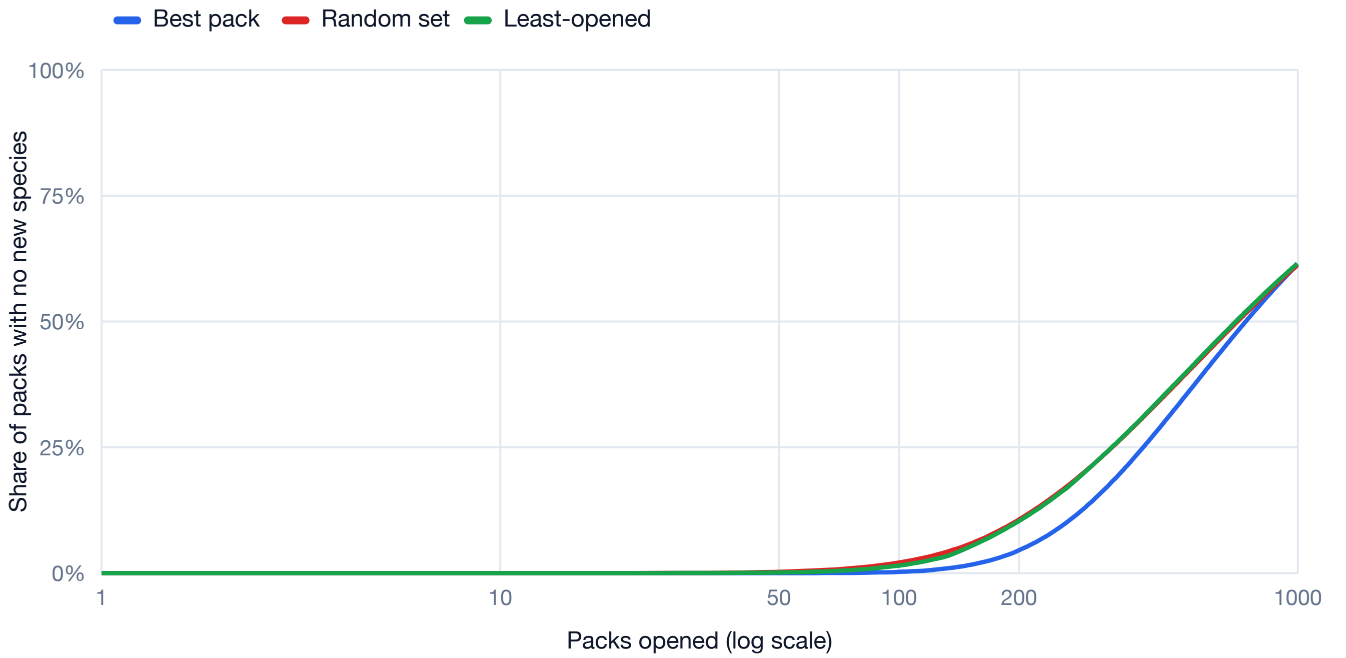

Figure A4. Share of packs that yielded *no* new species. Random and least-opened waste packs much sooner because they keep buying sets whose Pokémon you already have.

By **1000 packs** best-pack reaches **1020** of the 1025 achievable species, versus **950** for random — and it gets there having wasted far fewer packs along the way.

Appendix — What you actually pull (all sets)

Following the best-pack strategy for 1,000 packs (averaged over 1,000 runs), you open about **10198 cards**: **8644 Pokémon** (85%), 1406 Trainers and 149 Energy. Those Pokémon cards resolve to **1020 distinct species**, which means roughly **88% of the Pokémon cards are duplicates** of species you already had — the price of chasing the long tail.

Cards by rarity (cumulative, average)

Rarity	@ 10	@ 50	@ 200	@ 1000
Common	**58**	**269**	**1046**	**5261**
Uncommon	**34**	**178**	**717**	**3578**
Rare	**10.0**	**49**	**199**	**1027**
Double Rare	**1.7**	**8.7**	**37**	**197**
Ultra Rare	**0.9**	**3.9**	**15**	**78**
Illustration Rare	**0.1**	**1.9**	**8.2**	**33**
Special Illus. Rare	**0.1**	**0.7**	**3.4**	**14**
Hyper Rare	**0.1**	**0.6**	**2.1**	**11**
All cards	**106**	**511**	**2026**	**10198**

Commons and Uncommons dominate every pack; the higher tiers come almost entirely from the single guaranteed "rare slot" (and the occasional reverse-holo).

What lands in the rare slot?

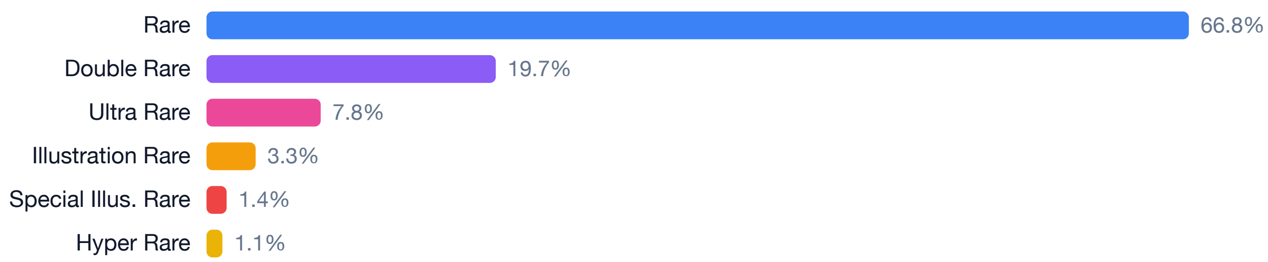

Figure A5. Of the one guaranteed rare-slot card per pack, how often each tier appears. Across all eras the mix skews further to plain Rare — older sets had few chase tiers, so those slots fall back to a Rare.

The most-pulled Pokémon

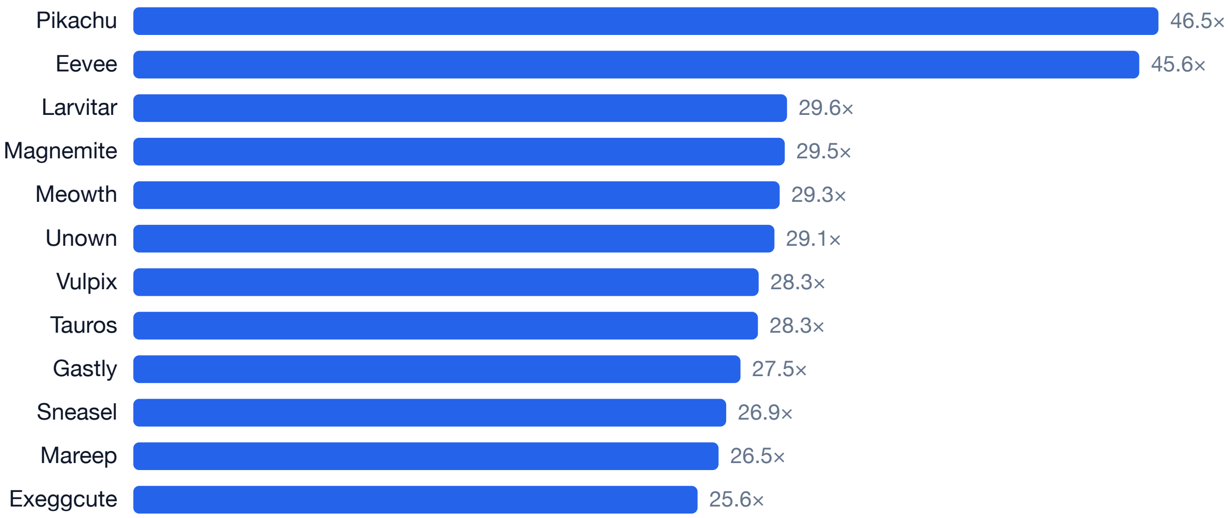

Figure A6. Average copies collected in 1,000 packs. These are common Pokémon reprinted across many of the sets best-pack opens — you end up swimming in them.

The hardest to pull

The 15 scarcest species that still showed up at least once — typically found only at high rarity or in sets best-pack rarely needs to open (every one of the 1025 reachable species turned up at least once).

Species	Gen	Copies	Species	Gen	Copies
Xurkitree	Alola	**0.94**	**Baxcalibur**	Paldea	**1.19**
Stakataka	Alola	**0.95**	**Spectrier**	Galar	**1.21**
Celesteela	Alola	**1.03**	**Runerigus**	Galar	**1.26**
Arctozolt	Galar	**1.09**	**Pheromosa**	Alola	**1.29**
Cursola	Galar	**1.10**	**Nihilego**	Alola	**1.30**
Toucannon	Alola	**1.11**	**Politoed**	Johto	**1.33**
Blacephalon	Alola	**1.16**	**Dracozolt**	Galar	**1.36**
Dracovish	Galar	**1.17**			

How much to actually complete the Pokédex?

Completing the binder means owning all **1025** species — which needs the full 133-set catalogue (the Scarlet & Violet + Mega sets alone top out at 933, so they can't finish it). Each **pack costs €10**; **singles cost €0.50** each (your "€5 for 10"), but a card can only be bought as a single if it's printed at an allowed rarity.

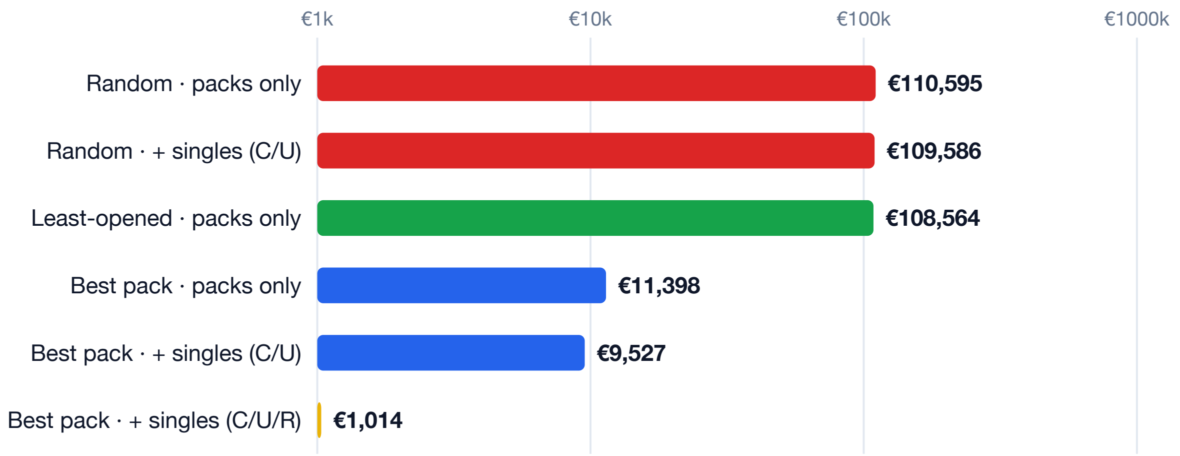

Figure 7. Expected total spend to complete the Pokédex (log scale, 500 runs each). Buying singles only helps if you can buy the bottleneck species.

Packs only: with the smart best-pack order you finish for about **€11,398**; mindless random/least-opened buying balloons to **€110,595+** — roughly 10× more — because they waste packs chasing the rare tail.

The Common/Uncommon wall. 125 of the 1025 species *never* appear as a Common or Uncommon in any set — they exist only at Rare or higher, and they are exactly the hardest to pull. Under a C/U-only rule you can't buy them as singles, so they must be pulled no matter what. That's why adding C/U singles barely dents random's bill (still €109,586): the unbuyable tail gates completion regardless of how many cheap commons you mop up.

Allow Rares as singles and the wall disappears. All **1025** species become buyable (0 pack-only), so you simply *buy* the dex — about 673 singles plus only ~68 packs — for roughly **€1,014**. At that point the pack strategy stops mattering: best, random and least-opened all finish near €1,000, because you're no longer relying on luck to find anything.

Strategy	Singles rule	Packs	Singles	Expected cost
Best pack	packs only	1,140	0	€11,398
	+ singles (C/U)	923	595	€9,527
	+ singles (C/U/R)	68	673	€1,014
Random	packs only	11,059	0	€110,595
	+ singles (C/U)	10,926	643	€109,586
	+ singles (C/U/R)	73	726	€1,094
Least-opened	packs only	10,856	0	€108,564
	+ singles (C/U)	10,677	638	€107,085
	+ singles (C/U/R)	72	720	€1,085

All singles priced at €0.50 including Rares. Bulk Rares cost a little more in practice; even at €2 per Rare-single the C/U/R total only rises to ~€1,300 — still an order of magnitude below grinding packs. Figures are Monte-Carlo means over 500 completions; random/least packs-only have wide run-to-run spread.

How long — and how cheaply — at one pack and ten cards a week?

Now make it a routine: **one Scarlet & Violet + Mega Evolution booster** (€10) plus **ten cards a week** through the "single" channel (your "€5 for 10"). For each of those ten you first try to **trade a spare you already own** — a duplicate of a species you've found — for a card you still need, and only pay €0.50 if you have no matching spare. Trades swap **like-for-like rarity** (a spare Common for a Common you're missing). Crucially, the singles and trades can come from **any set in the catalogue**; only the booster is restricted to Scarlet & Violet + Mega.

> **About 72 weeks — roughly 1.4 years**
>
> And it barely wavers: the median is 72 weeks, with the middle 80% of runs between 72 and 73. There's almost no luck left — once you can *buy or trade* any species, finishing is just arithmetic.

The pace is set by the **ten-cards-a-week** rate, not by pack luck. Of the 1025 species, about **720 come through that weekly channel** (≈ 72 weeks × 10) and the **~72 boosters** you open along the way grab the other **~305 for free**. Ten a week is ~520 a year, so a little over two years of singles alone — the incidental pack pulls shave it to ~1.4.

Trading doesn't make it faster — it makes it cheaper. Whether each weekly card is a trade or a purchase, you still add ten a week, so completion lands at the same ~72 weeks either way. What changes is the bill.

Figure 8. Total spend to finish at this weekly pace (500 runs). The ~€724 of boosters is a fixed floor; trading your spare commons nearly halves the singles bill on top.

Of the ~720 cards you pull through the weekly channel, about **301 (42%) can be covered by trading a duplicate** — almost all of them commons you opened more than once — so you only **buy ~419**. That nearly halves the singles bill and trims the total from **€1,084** to **€933**, a saving of about **€150**.

Plan	Time	Packs	Cards traded	Cards bought	Total
Buy every card	**72** ~1.4 yr	**72** €724	**0**	**720** €360	**€1,084**
Trade spares first	**72** ~1.4 yr	**72** €724	**301**	**419** €210	**€933**

Why trades cap out around 301. Your spares are overwhelmingly commons, and a common spare can only be swapped for a card printed as a common *somewhere*. The hardest missing species are rare-only, so a pile of common spares can't reach them — those you still have to buy. **You also can't finish on Common/Uncommon alone:** 43 of the 1025 species appear neither as a C/U single in any set nor in a Scarlet & Violet + Mega booster, so a C/U-only plan stalls at 982/1025. Allowing Rares (to buy or trade) closes the gap to all 1025.

Reproduce: `npx tsx scripts/sim/run.ts --trials 2000 --seed 1234 --budget 1000` · setPools hash e1d87a43d2902f40 · Node v24.14.1 · run took 75.2s · checkpoints [10, 50, 200, 1000]